



Published in final edited form as:

Comput Biol Med. 2020 December ; 127: 104099. doi:10.1016/j.compbiomed.2020.104099.

The Effect of Airway Motion and Breathing Phase During Imaging on CFD Simulations of Respiratory Airflow

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Abstract

Rationale: Computational fluid dynamics (CFD) simulations of respiratory airflow can quantify clinically useful information that cannot be obtained directly, such as the work of breathing (WOB), resistance to airflow, and pressure loss. However, patient-specific CFD simulations are often based on medical imaging that does not capture airway motion and thus may not represent true physiology, directly affecting those measurements.

Objectives: To quantify the variation of respiratory airflow metrics obtained from static models of airway anatomy at several respiratory phases, temporally averaged airway anatomies, and dynamic models that incorporate physiological motion.

Methods: Neonatal airway images were acquired during free-breathing using 3D high-resolution MRI and reconstructed at several respiratory phases in two healthy subjects and two with airway disease (tracheomalacia). For each subject, five static (end expiration, peak inspiration, end inspiration, peak expiration, averaged) and one dynamic CFD simulations were performed. WOB, airway resistance, and pressure loss across the trachea were obtained for each static simulation and compared with the dynamic simulation results.

Results: Large differences were found in the airflow variables between the static simulations at various respiratory phases and the dynamic simulation. Depending on the static airway model

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Conflict of Interest: Dr. Bates has research agreement with Siemens PLM software.

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used, WOB, resistance, and pressure loss varied up to 237%, 200%, and 94% compared to the dynamic simulation respectively.

Conclusions: Changes in tracheal size and shape throughout the breathing cycle directly affect respiratory airflow dynamics and breathing effort. Simulations incorporating realistic airway wall dynamics most closely represent airway physiology; if limited to static simulations, the airway geometry must be obtained during the respiratory phase of interest for a given pathology.

Keywords

Airway CFD; Respiratory Airflow; Airway Motion; UTE MRI; Unsteady; Trachea

1. Introduction

Computational Fluid Dynamics (CFD) simulations of human respiratory airflow reveal the relationship between airway anatomy and airflow in health and disease, providing clinically relevant information that cannot be determined *in vivo*. For example, CFD simulations are capable of quantifying resistance to airflow, pressure losses, and respiratory effort [1–7]. The clinical impact of this technique has been demonstrated through deposition mapping of inhaled therapeutic particles and improved surgical planning for conditions with abnormal airway anatomy [8–10]. Most studies of airflow are based on anatomical airway models obtained from clinical imaging, usually computed tomography (CT) [4,11–13] or magnetic resonance imaging (MRI) [2,3,14], with virtual airway surfaces generated from segmentation of these anatomical images. Such airway surfaces act as anatomical models for both virtual CFD simulations and fabricated airways in physical *in vitro* experiments [3,15].

These images are typically static and therefore the airway geometries used in studies are usually also static too. However, the airway changes in size and shape during different breathing maneuvers, including restful breathing [16]. Furthermore, as clinical images are often obtained retrospectively from clinical databases, the phase of breathing during which images were captured is not always reported [12,17].

CFD studies that do not incorporate motion in the airway model often perform steady flow simulations at a particular period of interest occurring during the breath, for example peak inspiration [1,13]. However, the images from which these studies obtain their geometries may not have been obtained during the same phase of breathing. For example, in some studies the images were obtained during a breath hold at total lung capacity, or at end expiration (either residual volume or functional residual capacity) [4,18,19]. Other studies do not report the phase of breathing in which the images were obtained or, in the case of MRI image acquisition, image acquisition may occur across many breaths producing an image of the average position of the airway [12,17]. As such, most CFD simulations of airflow in the central airways do not sufficiently account for the dynamic changes in airway size and shape throughout a breath.

The effect of the natural motion of the airway as a geometric boundary condition has not been frequently studied. Cherobin et al. [20] showed that small changes in nasal geometry can change resistance by at least 35%. A study showed that pressure drop from the top of the

trachea to the distal ends of 52 branches of the airway at peak inspiration was changed by 24% in dynamic and static airway models [21]. Bates et al. [22] showed in one case that pressure loss changed by 14.6% during inspiration and by 19.2% during expiration from the nose (including a mask) to carina in a model with motion vs. a static model. However, that was during a forced breathing maneuver, and the effects during restful breathing were not investigated.

The accuracy of results obtained from CFD simulations depends on the accuracy and physiological relevance of the boundary conditions. Other boundary conditions, such as the inflow and outflow boundary condition choices have received significant attention in the literature. For example, Taylor et al. [23] have compared the outcomes of CFD simulations due to different inflow boundary conditions. Parameters regarding how the flow-governing Navier-Stokes equations are solved have also received significant attention. For instance, several studies have compared the use of different turbulence models [15,24–26].

The goal of this study is to investigate the effect of airway motion on clinically relevant airflow measures by comparing a CFD simulation with prescribed realistic airway wall motion and five static CFD simulations (four static CFD simulations using airway surfaces obtained at four different phases of breathing, and one static CFD simulation using the ungated airway which was an average image taken over many breaths during the scan) for each subject. We examined two healthy subjects and two subjects with dynamic tracheal collapse (tracheomalacia, TM) as a paradigmatic airway disease. In order to determine the effect of using retrospectively obtained static airway images, this study also compares the differences among static airway models based on images obtained at various phases of breathing and the ungated, to guide future investigators who may not have access to dynamic imaging. This comparison demonstrates how these changes in anatomy affect CFD-derived measurements such as pressure drop, WOB, and resistance in the trachea.

2. Methods

2.1. Study Subjects

Four neonatal intensive care unit (NICU) subjects from Cincinnati Children's Hospital were recruited with authorization from the Institutional Review Board and written parental consent. All subjects were not intubated at the time of MRI and were freely breathing room air, except one subject who received non-invasive respiratory support via high flow nasal cannula (HFNC). Two subjects were diagnosed with TM via clinical bronchoscopy and had comorbid lung disease of prematurity (bronchopulmonary dysplasia, BPD), and the other two subjects were controls who had non-respiratory conditions but no airway issues. All neonates were imaged at postmenstrual age of 41.6 ± 1.8 weeks.

2.2. Airway Imaging

Ultrashort echo time (UTE) MRI is a free-breathing, non-ionizing imaging technique that can be applied to capture 3D high-resolution, dynamic motion of the airway particularly in the neonatal population [6,27–31]. UTE MR images were acquired using a neonatal-sized 1.5 T scanner sited within the NICU (orthopedic scanner from ONI Medical systems

configured for pediatric imaging; currently operating with GE Healthcare software) with imaging parameters shown in Table 1 [30,32–36]. Image coverage was from the nasopharynx through the base of the lungs in order to capture both airway and lung dynamics.

2.3. Reconstruction of Images at Different Phases of Breathing

Previous studies have demonstrated that UTE MR images acquired during tidal breathing can be reconstructed retrospectively at different points in the breathing cycle [6,29–31]. Briefly, the initial point of the free induction decay (FID, magnitude and phase) signal of MRI was used to generate a respiratory waveform, which was modulated due to breathing and bulk motion of the neonate. The bulk motion can be distinguished and removed from the respiratory waveform. The respiratory waveform was gated into four bins by using the amplitude of each breathing cycle after discarding the bulk motion, if present. The four bins correspond to UTE MRI data acquired during end expiration, peak inspiration, end inspiration, and peak expiration (Figure 1). For each data bin, an image was reconstructed which shows the specified phase of breathing [14,30,32,37,38]. Data acquired in between these bins were discarded to reduce motion blurring in the images (shown in grey, Figure 1). The ungated airway image was reconstructed without using any bin assignment (gating) and the entire FID waveform was used to obtain the image.

2.4. Dynamic Airway Surfaces and Airflow Rates for CFD Boundary Conditions

Airway surfaces were segmented from each MR image via ITK-SNAP software (Penn Image Computing and Science Laboratory, USA, version 3.8.0) using a 3D active contour segmentation method [39]. The average intensities of the air-filled airway and the neighboring soft tissue were used to calculate an intensity threshold for segmentation [31]. Surfaces were smoothed using a Taubin filter (smoothing parameters, $\lambda = 0.6$ and $\mu = -0.6$) in MeshLab 2016 (Visual Computing Lab, Italy) [40,41]. All segmented airway surfaces extended from the nasopharynx through the main bronchi. To obtain the motion of the airway between images (temporal resolution of 0.8 ms was required for CFD simulations), surface registration was performed as described by Bates et al. [14] using MIRTk 1.1 (<https://mirtk.github.io/> – Department of Computing, Imperial College London, UK). Surface registration maps all nodes on the airway surface at the initial image time-point to the airway surfaces at each subsequent time-point. CFD simulations require a much finer temporal resolution than the imaging temporal resolution [0.8 ms vs 0.22 s]. To determine the position of the nodes on the airway wall in-between the image time-points, the nodal positions were interpolated using the cubic b-spline function. This study used the surface registration and interpolation techniques described previously by Bates et al. [14].

A median respiratory waveform for one breath was calculated from the entire respiratory waveform (~1000 breaths per scan) to obtain a representative breathing profile for each subject. The median waveform was then scaled according to the left and right lung tidal volumes, which were generated via calculating the difference in segmented lung volumes between the end inspiration and end expiration MR images (Figure 2A) [30,42,43]. This volume-time waveform was then differentiated with respect to time to produce a flowrate-time curve (Figure 2B).

2.5. CFD Mesh Generation and Physical Models

Unsteady CFD simulations were performed using the commercial CFD package, STAR-CCM+ 14.04.011-R8 (Siemens PLM Software, Plano, TX, USA). The discretized airway meshes included approximately 2 million cells per airway model [6]. Meshes consisted of polyhedral cells in the bulk of the airway and nine prism layers on the walls. A grid independence study was performed using five different mesh sizes (0.5 Million (M), 1 M, 2 M, 5 M, and 10 M) prior to the simulations. The percentage change in the pressure loss across the trachea at peak inspiration was 0.65% between the 2 M and 10 M simulations. Air was considered incompressible. The Large-Eddy Simulation (LES) turbulence model was utilized to resolve large scale turbulence and model the effect of subgrid-scale turbulence. The wall-adapting local eddy-viscosity (WALE) subgrid-scale model was implemented to describe the flow near the walls and to resolve small scale turbulence [44]. It was assumed that at the wall of the airway, airflow velocity was zero relative to the airway wall (no-slip condition). Unsteady CFD simulations had second-order temporal discretization with 0.8 ms time step. A time step convergence study was also performed using four different time steps (0.01 ms, 0.08 ms, 0.8 ms, and 2 ms). The percentage change in the tracheal pressure loss at peak inspiration was 0.62% between 0.01 ms and 0.8 ms simulations. To ensure that the finest simulation (10 million cells and time step of 0.01 ms) acted as a suitable reference simulation, the Kolmogorov length scale and time scale was calculated for this finest simulation, as previously described [45,46]. The mean ratio of the time step of the simulation and the Kolmogorov time scale was 0.025. The ratio of the length of the cell size and the Kolmogorov length scale was 0.55. Therefore, over the vast majority of the geometry and breath, this reference simulation is a Direct Numerical Simulation (DNS). Since the less finely resolved simulations produce extremely similar results to this high-resolution simulation, as stated above, the mesh and temporal discretizations were considered adequate for this study. The surface node positions at each time-step were used to morph the surface of the CFD mesh to a new position for each time-step. The nodes of the interior volume mesh were also moved via a cubic b-spline interpolation function based on the surface motion [22]. The flow-governing Navier-Stokes equations were modified to incorporate the motion of the mesh as previously described [22] and were solved using a segregated flow solver.

The airflow pressure was assumed to be at atmospheric pressure at the inlet, neglecting pressure losses in the nose. The relative pressure drop along the trachea – the specific area of interest – is not affected by small variations in the absolute pressure upstream of the trachea. The larynx, pharynx, and bronchi are included to provide realistic flow patterns entering the trachea during inspiration and expiration. Since the region of interest is the trachea, the addition of the upper airway and main bronchi to the virtual airway allows turbulence to develop naturally before flowing into the trachea [47]. Meshes were extruded from the inlet and outlet planes to overcome flow instability at the boundaries due to backflow [48]. A CFD simulation was performed for each subject with 6 models: one with a moving airway, four with a static airway derived from various phases of breathing (end expiration, peak inspiration, end inspiration, and peak expiration), and one using the ungated airway. All simulations had transient inflow conditions representing the full breathing cycle. On average, CFD run time was around 720 core-hours.

2.6 Airway Geometric and Flow Measurements

Cross-sectional planes were created along the airway using a centerline to calculate the cross-sectional area (CSA) of the airway from nasopharynx to carina as described previously [12,45,49]. For the dynamic case, these measurements were repeated throughout the breath. To calculate the Reynolds number along the airway, the hydraulic diameter (D) of each plane was measured and the surface averaged velocity of each plane was evaluated using the CFD simulations [9]. D is calculated by the following: $D = 4 \cdot \text{CSA} / P$, where P is the perimeter of the plane.

The component of WOB due to resistance to airflow was calculated for each simulation as described by Bates et al. [5]. To calculate the tracheal resistive WOB per day, first the instantaneous energy flux (Φ_E) at planes at either end of the trachea (specifically, the glottis, carina) was calculated using the following equation.

$\Phi_E = \oint_A P_T (\mathbf{U} \cdot \hat{n}_A) dA$ Where P_T is the total pressure, $\mathbf{U}$ is the mean velocity vector, and $\hat{n}_A$ is the normal vector at each plane [5].

The instantaneous energy flux difference between the two planes was calculated throughout the breathing cycle. These values were then integrated with respect to time to obtain the tracheal resistive WOB per breath. This value multiplied by the number of breaths per day (calculated from the respiratory rate) gave the total tracheal resistive WOB per day.

Airway resistance and the pressure drop in the trachea were also compared with the dynamic CFD results. Airway resistance was calculated instantaneously throughout the breathing cycle and then integrated with respect to time over the duration of one breath in order to compare cases. Tracheal resistance, R, is calculated: $R = \Delta P / Q$, where ΔP is the pressure drop across the trachea (mean total pressure on a cross-sectional plane at the glottis minus mean total pressure on a cross-sectional plane at the carina) and Q is the volumetric flow rate through the trachea.

3. Results

3.1 Changes in Airway Geometries throughout the Breath

Figure 3 shows the CSA along the airway from the nasopharynx to carina of the four subjects. In all airway geometries, CSA was largest at end inspiration in the majority of the airway. The end inspiration airway had the highest airway volume from the nasopharynx to carina for each subject. In the healthy subjects (Subjects 1 and 2), significant variation in CSA was observed throughout the breath with mean variations in CSA of 30% for Subject 1 and 24% for Subject 2 from the end expiration time-point, with some regions varying by up to 68% (Figure 3A and 3B). The mean change in CSA along the trachea from the end expiration time-point to the time-point of most difference in subjects 3 and 4 was 91% and 27%, respectively. The CSA of Subject 3 was increased by more than 100% after 50 mm from the nasopharynx to carina at end inspiration compared to the end expiration (Figure 3C). The CSA of the glottis of the subject 4 diagnosed with TM changed from 2.6 mm² to 16.6 mm² (Figure 3D). Furthermore, in the lower region of the trachea (more than 40 mm from nasopharynx), the location with the highest change in CSA expanded from 10.7 mm²

to 21.8 mm² during the breathing cycle, an increase of 104%. The CSA of the ungated airway in all four subjects was approximately in the middle of the dynamic region.

3.2 Flow Rate Analysis

The lung tidal volumes of the four subjects were derived from changes in lung volume between the different phases of breathing. The lung tidal volume and the ratio of the right to left lung tidal volumes, and the respiratory rates of the four subjects are given in Table 2. Results show that the right lung tidal volume of the subject 3 diagnosed with TM is almost twice that of the left lung tidal volume.

3.3 Variation of the Reynolds Number along the Airway

For each airway, the mean cross-sectional Reynolds number along the airway was calculated at peak inspiration and peak expiration of the airflow for the two example subjects with and without TM. In the subject without TM (Subject 2), the variation in the Reynolds numbers obtained in different airway models was minor at peak inspiration, except at around 45 mm from the nasopharynx (Figure 4A). However, in the subject with TM (Subject 3) at peak inspiration, the Reynolds number fluctuates throughout the airway (Figure 4C). The largest variation in Reynolds number due to different geometries was found approximately 12 mm from the nasopharynx where the Reynolds number changed from approximately 1000 (end inspiration geometry) to 3500 (dynamic airway). Similar variations can be observed for each subject at peak expiration.

3.4 The Effect of Airway Geometry on Work of Breathing, Airway Resistance, and Pressure Drop

To compare the effect of using different airway geometries when performing CFD simulations, tracheal resistive WOB per day, tracheal airway resistance integrated with respect to time over one breath, and the pressure drop across the trachea at peak expiration were calculated for each subject. All airflow measurements (WOB, resistance, and pressure) were compared with the respective dynamic airway CFD simulation results from each subject. The tracheal resistive WOB per day is shown for each subject and geometry in Figure 5A. Figure 5B illustrates the percentage differences in tracheal resistive WOB compared to dynamic airway WOB for the four subjects at each breathing phase and using the ungated airway. In Subject 1 (without TM), tracheal resistive WOB per day calculated using the end expiration geometry was over 100% higher than the value calculated in the dynamic airway. In Subject 3 (with TM), tracheal resistive WOB was 82% lower in the end inspiration geometry compared to the dynamic airway. Subjects 2 and 4 also had similar differences in these metrics when using various geometries. Figure 5 shows the large variations in tracheal resistive WOB depending on the initial surface used. Specifically, Figure 5B demonstrates that no one particular time-point best represents the dynamic airway in all the subjects and each of the four subjects has a different time-point that is least representative of the dynamic airway. Furthermore, tracheal resistive WOB calculated using the ungated airway indicates that it does not produce the same airflow measurements as the dynamic airway. In the subjects with TM, (subjects 3 and 4) the tracheal resistive WOB in the ungated airway was 50% lower than the dynamic case on average. In the subjects without TM, the tracheal resistive WOB in the ungated airway of subject 2 was also lower

than the dynamic case, but in subject 1 the value was higher than the dynamic case, but the differences were smaller (23% and 6% respectively). Full results for all variables are given in Appendix A (Table A).

Analysis of tracheal airway resistance integrated with respect to time over one breath produced similar variation compared to WOB. For example, the integrated resistance of Subject 1 (without TM) was 105% higher in the end expiration geometry than the dynamic case. In the two subjects with TM (Subjects 3 and 4), pressure drop across the trachea at peak expiration of the airflow was lower in the end inspiration geometry by more than 85% compared to the dynamic airway of each subject. Furthermore, pressure drop across the trachea of Subject 4 (with TM) calculated using any static geometry including the ungated airway was more than 38% lower compared to the dynamic airway of that subject.

3.5 Velocity Distribution of the Airway Geometries

To identify the flow mechanisms that cause the differences in static and dynamic airway CFD simulation results, the velocity distribution at peak inspiration of the airflow was compared between the dynamic and static airways (Figure 6). A strong jet can be observed at the glottis, carina, and mid-trachea for the end expiration and peak expiration static airway geometries. In contrast, much weaker flow can be observed throughout the end inspiration static airway. However, velocity of the flow in the dynamic airway was lower compared to the end expiration and peak expiration airways and higher compared to the end inspiration and the ungated airway. Furthermore, similar airflow velocities can be observed throughout the dynamic airway and the peak inspiration airway except in the glottis region.

4. Discussion

Geometric analysis of the trachea throughout a breathing cycle reveals significant variation in tracheal cross-sectional area and volume in healthy subjects and those with an airway disease such as, TM. Figure 3D shows this variation is not uniform along the length of the airway, with some sections relatively static (e.g. the cricoid cartilage at approximately 25 mm along the airway) and some highly mobile (e.g. the glottis at approximately 18 mm). Szelloe et al. [50] show that in the 0–1 age group median CSA of the upper to mid trachea is 33 mm^2 ranging from $17.4 - 55.8 \text{ mm}^2$. Our results fall within the range of the above findings. The motion of the glottis is particularly important, as the opening between the vocal folds is often the narrowest point in the airway. This point is causing the highest local resistance and dominates the inspiratory flow in the trachea by causing the formation of the laryngeal jet [51,52]. Therefore, the large fluctuations in CSA at the glottis can have a large impact on respiratory aerodynamics, as this study has demonstrated (Figure 6). The mean Reynolds number along the airway of the same subject (Subject 3) is plotted in Figure 4C. The large variation in CSA at the nasopharynx is revealed by the large increase in Reynolds number at the nasopharynx between the same flowrate moving through geometries obtained at end inspiration and peak inspiration. The significance of the Reynolds number is due to its relationship to the turbulence of a flow. In steady flows within long, straight, smooth pipes begin transition to turbulence at Reynolds numbers of approximately 2300. Although none of these assumptions are true in respiratory flow, the Reynolds number of a given flow is

often used to justify the choice of turbulence models used [9,53]. Turbulent, transitional, and laminar flows will result in different airway resistance values and will have a significant effect on clinically important factors such as particle deposition, due to the different mixing characteristics. Therefore, if airflow is modeled in a geometry produced during a phase of breathing that results in laminar flow, very different flow characteristics may be observed than in a geometry that causes more turbulence in the flow. This difference may limit the reliability of CFD to aid clinical decisions in cases where the geometry cannot be matched to the phase of breathing or where dynamic simulations cannot be used, as demonstrated in this study.

Figure 5 shows the effect of the time-point during the breathing cycle in which geometry was obtained on tracheal resistive WOB. In healthy cases, a difference of up to 102% was observed between geometries obtained at different phases of breathing. Similarly, in subjects with TM, the difference was as high as 237%. Interestingly, Figure 5B shows that the biggest difference to the dynamic case was found at four different time-points for the four subjects indicating that there is no specific static airway surface which provides the same results as a dynamic airway. A typical MRI scan generates an ungated airway image which shows the average airway position over the many breathing cycles performed during the scan. However, Figure 5 demonstrates that the tracheal resistive WOB calculated using an airway model obtained from the ungated image was not closer to the value calculated from the dynamic airway CFD than gated images that represent the airway at a certain time-point. Figure 5B also shows that the percentage difference of the tracheal resistive WOB was higher in the two subjects with TM compared to the subjects without TM. Furthermore, the average tracheal resistive WOB of each subject using the four different phases of breathing was not equivalent to that value obtained using the ungated airway. For tracheal resistance integrated over one breath, the largest difference was 105% in controls and 200% in subjects with TM (Appendix A, Table A). In the case of tracheal pressure loss at peak expiration flow, the differences were 94% and 90%. These large disparities reveal the effect that temporal CSA variations have on respiratory airflow and demonstrate the importance of basing CFD simulations on airway anatomy at the same phase of breathing as the phase that will be simulated.

While these results show that CFD simulations of airflow at a single time-point (i.e. steady simulations) produce different results depending on the time-point at which the geometry was acquired, Figure 6 reveals that even when the surface and airflow time-point match, the results may still differ from a fully dynamic simulation of a full respiratory cycle. While conservation of mass requires that the average velocity must be the same through the same cross-section, the velocity distribution maps are quite different (compare glottis planes in Figure 6B and 6F). The difference between pressure loss at peak inspiration under static and dynamic geometries was as high as 19%, even though the geometries were instantaneously identical at that time-point. The likely reason for this is the dependence of flow not only on current conditions, but on the flow at prior time-points (i.e. the **transient** term in the flow-governing Navier-Stokes Equations).

Respiratory CFD simulations have demonstrated the potential for use in clinical settings, for example, to determine the effect of disease on patient wellbeing, to understand the impact of

specific surgical interventions, and to predict the deposition of inhaled drugs. The results of this study show the importance of incorporating physiological airway motion into CFD studies. While we demonstrated that incorporating this motion was more important in subjects with a dynamic airway disease such as tracheomalacia, significant airway motion was also observed in healthy subjects, indicating that airway motion should be considered for airway studies in any population. The results reported in this study are all derived from CFD simulations, though the concepts investigated also hold true for *in vitro* experimental modeling of respiratory aerodynamics based on medical imaging.

The value of patient-specific airway modeling is to provide insight into a specific subject's airway disease and to regionally assess where losses occur. This is particularly relevant in subjects with multi-level obstruction (e.g. TM and subglottic stenosis) or with airway obstruction and lung disease. In these cases, the clinical question is which abnormality should be treated to best alleviate symptoms. The results of this study show that in order to accurately assess an individual's airway, the motion of the airway must be taken into account. The additional computational cost of the dynamic simulations compared to static cases was approximately a 25% increase in core-hours to achieve a solution. Given the large discrepancies between static and dynamic results, this increase in CFD run-time may often be considered worthwhile to produce realistic patient-specific results.

Previous studies have compared particle deposition of the lungs and the pressure loss of the airway in static and dynamic CFD simulations. Miyawaki et al.[54] have shown in a study that static vs. dynamic imaging also affects particle deposition in the lungs. They found that 18% more particles entered the right middle lobe for the static case compared to the dynamic case using adult subjects. Another study showed that pressure loss to the carina varied by 26.9% compared to the dynamic case when using a static geometry at end expiration in an adult [14]. These studies also emphasize the importance of the boundary conditions (i.e. static vs. dynamic) in the airway as the outcomes clearly depend on the presence of motion. Another study showed that on average, the effect of airway motion in subjects with TM is a 337% increase in the energy required for breathing, whereas this increase is only 24% in subjects without TM [6]. Previous studies have quantified resistive WOB of the respiratory system in neonates by measuring the lung tidal volumes and esophageal pressure [55–57] or transpulmonary pressure [58]. Esophageal pressure was measured as a substitute for the pleural pressure. These studies have reported values of energy expenditure ranging from 89.28 J/day to 616.32 J/day for healthy neonates and neonates with different lung diseases such as respiratory distress syndrome [6]. In the present study, the value for tracheal resistive WOB alone ranged from 51.9 J/day to 635.5 J/day demonstrating that the amount of energy needed to move air through the trachea can reach the same order of magnitude as that for the entire respiratory system, particularly in neonates with TM. Previous techniques were not able to quantify the WOB based on the regions of the airway. However, our method can be used to quantify each part of the airway separately and identify the region that contributes the most to losses. The imaging and flow modeling techniques used in this study are also applicable to older infants and adults. These older populations can also suffer from diseases that result in dynamic airway narrowing and are known as tracheomalacia or excessive dynamic airway collapse (EDAC) depending on the cause of motion.

The limitations of this study include the image resolution. The UTE MRI reconstruction used in this study produces $0.7 \times 0.7 \times 0.7$ -mm voxels, which is therefore the limitation of the spatial resolution of our airway reconstructions and motion. The temporal resolution of the MRI was one quarter of the breathing cycle duration (approximately 0.22 s in this population) and the acquisition window was $1/8^{\text{th}}$ of the breathing cycle duration (0.11 s). Motion occurring faster than this resolution was not captured. Another limitation of this study was using one breathing cycle to evaluate respiratory airflow measurements, rather than multiple cycles. A static simulation (end expiration geometry) and a dynamic simulation with two breathing cycles were performed to determine the effect of neglecting cyclic effects. The percentage difference of the tracheal pressure loss at peak expiration of the first breath and the second breath was 0.79% in the static simulation and 0.27% in the dynamic simulation. The techniques used in this study can be applied to calculate the CFD boundary conditions from various imaging modalities that can capture the dynamics of the airway lumen, including UTE MRI, dynamic MRI, or cine CT. Cine CT produces better spatial and temporal resolution than the UTE MR images used in this study, but has the disadvantage of exposing subjects to ionizing radiation. Thus, the MRI-based methods we utilize in this study can be more readily implemented in pediatric and/or radio-sensitive populations, such as the neonatal patients included here. Neonates were chosen for this study due to the high prevalence of TM in this population [29,59]. However, the results and the methodology are applicable to older children and adults as demonstrated previously [14,22]. Future work will be focused on quantifying the airway differences at multiple phases of breathing in older children and adults.

5. Conclusion

This study demonstrates the importance of incorporating unsteady boundary conditions, such as tracheal motion and airflow rates through the breathing cycle when performing CFD simulations of respiratory airflow. The tracheal area varies considerably during the breathing cycle, particularly in the lower region of the trachea, which directly affects resulting respiratory airflow metrics; indeed, this study found that the work of breathing, airway resistance, and the pressure drop across the trachea differ significantly between dynamic and static airway simulations in health and disease. This study has demonstrated the importance of performing CFD simulations with physiologically realistic dynamic boundary conditions in order to yield clinically relevant results. In cases where dynamic imaging is not available, for examples where images are obtained retrospectively, static images should be obtained at the same phase of breathing as the phase which is to be modeled and this choice of images should be reported in future studies.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgments

The authors acknowledge the financial support from NIH K99 HL144822, NIH T32 HL007752 and NIH R01 HL146689.

Appendix A

Table A.

Comparison of CFD measurements with respect to the dynamic airway result

J/Day (% Difference to Dynamic Geometry Case)						
Geometry	End Expiration	Peak Inspiration	End Inspiration	Peak Expiration	Ungated	Dynamic
Subject 1 (No TM)	30.79 (102%)	15.36 (1%)	14.22 (-7%)	15.55 (2%)	16.24 (6%)	15.27
Subject 2 (No TM)	93.43 (-13%)	165.85 (54%)	89.38 (-17%)	85.59 (-20%)	82.80 (-23%)	107.35
Subject 3 (TM)	499.75 (-21%)	605.35 (-5%)	112.76 (-82%)	537.07 (-15%)	341 (-46%)	635.45
Subject 4 (TM)	52.81 (2%)	31.66 (-39%)	17.07 (-67%)	174.82 (237%)	24.94(-52%)	51.90
Tracheal Airway Resistance Integrated w.r.t. Time Across One Breathing Cycle						
Pa.s ² /ml (% Difference to Dynamic Geometry Case)						
Geometry	End Expiration	Peak Inspiration	End Inspiration	Peak Expiration	Ungated	Dynamic
Subject 1 (No TM)	0.41 (105%)	0.19 (-5%)	0.17 (-15%)	0.14 (-30%)	0.17 (-15%)	0.2
Subject 2 (No TM)	0.28 (-33%)	0.50 (19%)	0.27 (-36%)	0.28 (-33%)	0.21 (-50%)	0.42
Subject 3 (TM)	2.39 (-27%)	2.71 (-17%)	0.39 (-88%)	2.52 (-23%)	1.39 (-57%)	3.26
Subject 4 (TM)	0.17 (-48%)	0.08 (-76%)	0.04 (-88%)	0.99 (200%)	0.06 (-82%)	0.33
Pressure Loss Across Trachea at Peak Expiration Airflow						
cmH ₂ O (% Difference to Dynamic Geometry Case)						
Geometry	End Expiration	Peak Inspiration	End Inspiration	Peak Expiration	Ungated	Dynamic
Subject 1 (No TM)	-0.15 (88%)	-0.09 (13%)	-0.07 (-13%)	-0.07 (-13%)	-0.09 (13%)	-0.08
Subject 2 (No TM)	-0.31 (-3%)	-0.62 (94%)	-0.32 (0%)	-0.28 (-13%)	-0.26 (-19%)	-0.32
Subject 3 (TM)	-1.29 (-15%)	-1.30 (-15%)	-0.23 (-85%)	-1.28 (-16%)	-0.74 (-51%)	-1.52
Subject 4 (TM)	-0.14 (-71%)	-0.07 (-85%)	-0.05 (-90%)	-0.30 (-38%)	-0.06 (-88%)	-0.48

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HIGHLIGHTS

- Airway motion needs to be taken into account to perform respiratory airflow CFD simulations.
- The impact of the breathing phase (end expiration, peak inspiration, end inspiration, and peak expiration) of the airway on respiratory airflow metrics has been demonstrated.
- Work of breathing, airway resistance, and pressure loss of the trachea heavily depends on the phase of breathing regardless of the condition of the subject (healthy/ disease).

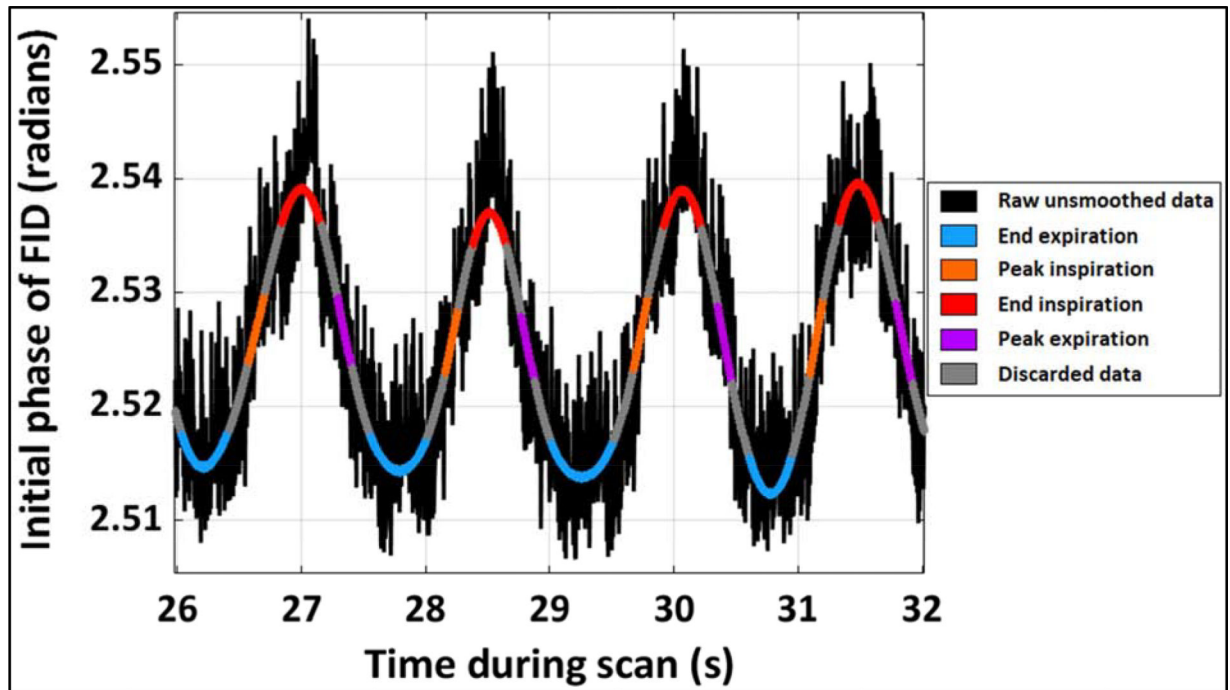


Figure 1.

The UTE-based respiratory waveform over a 6-second period of the neonatal MRI scan. Raw unsmoothed data (black) were smoothed and assigned to four respiratory bins (blue – end expiration, orange – peak inspiration, red – end inspiration, and purple – peak expiration). Each bin was used to create four distinct images from the breathing cycle. All data (four respiratory bins and the discarded data) was used to reconstruct the ungated airway image.

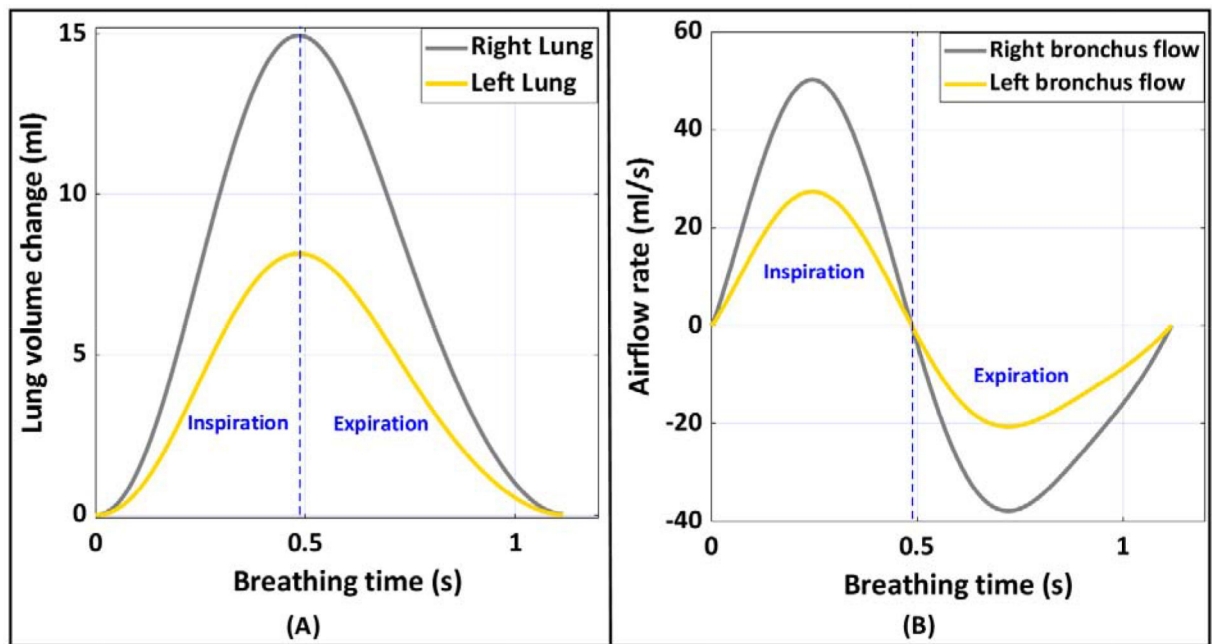


Figure 2.

Left and right lung volume changes during the breathing cycle (A) and the airflow rates of the left and right bronchus derived from the change in lung volumes of the subject with TM (subject 3) (B). The median respiratory waveform was rescaled according to the lung tidal volumes to obtain lung volume changes.

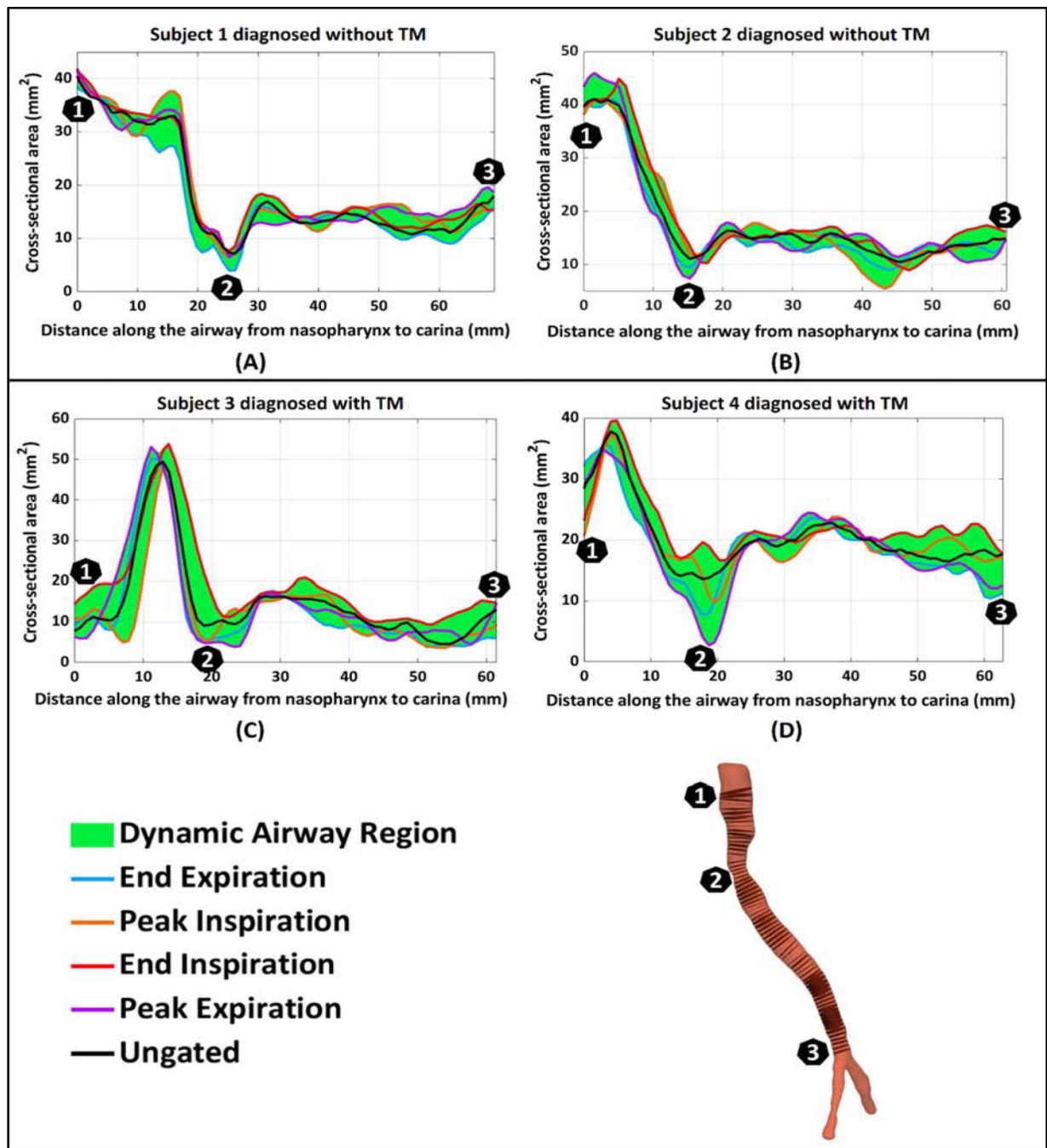


Figure 3.

The CSA of the (A) subject 1, (B) subject 2 diagnosed without TM and (C) subject 3, (D) subject 4 diagnosed with TM measured along the airway from the nasopharynx to carina. Numbers 1, 2, and 3 show the position of the nasopharynx, glottis, and the carina, respectively. The green colored region represents the range of airway motion at each point of the airway. The CSA of the end expiration, peak inspiration, end inspiration, peak expiration, and the ungated airways are shown in blue, orange, red, purple, and black respectively. The airway geometry at end inspiration had the largest CSA in most regions of

the airway. The CSA of the ungated airway was approximately in the middle of the range of CSA values covered by the dynamic airway during a breath.

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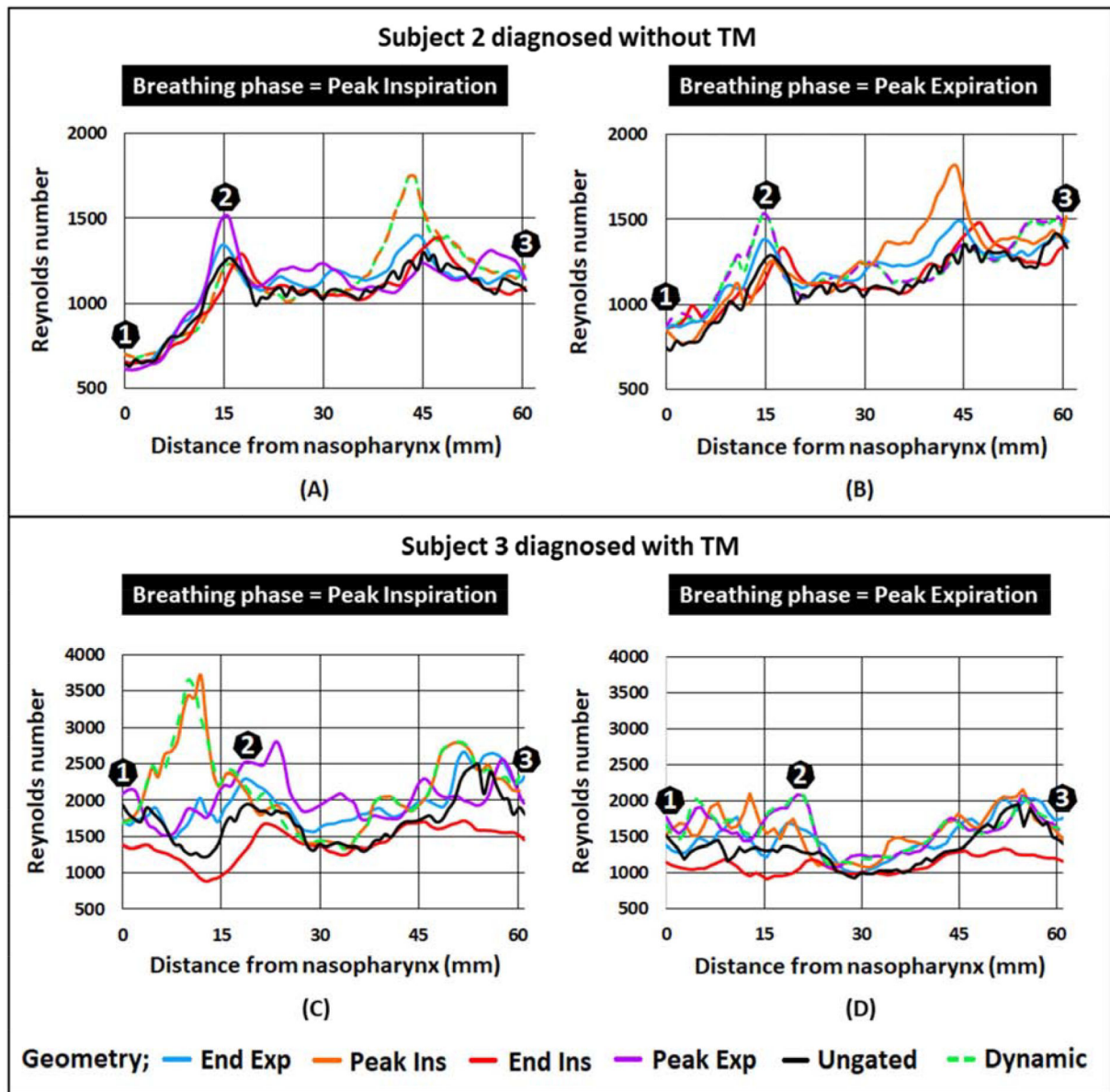


Figure 4.

Reynolds number along the airway for airway geometries at different phases of breathing (blue – end expiration, orange - peak inspiration, red – end inspiration, purple – peak expiration, black – ungated, and green - dynamic). (A) and (B) show the variation in the Reynolds number along the airway at peak inspiration and peak expiration, respectively, for an example subject without TM (Subject 2). (C) and (D) show similar information for an example subject with TM (Subject 3). The positions of the nasopharynx, glottis, and carina are indicated in numbers 1, 2, and 3 respectively. In the subject with TM, the fluctuation of the Reynolds number along the airway at peak inspiration is greater than in the subject without TM. The other two subjects (Subjects 1 and 4) displayed fluctuations in the Reynolds number similar to Subjects 2 and 3, respectively.

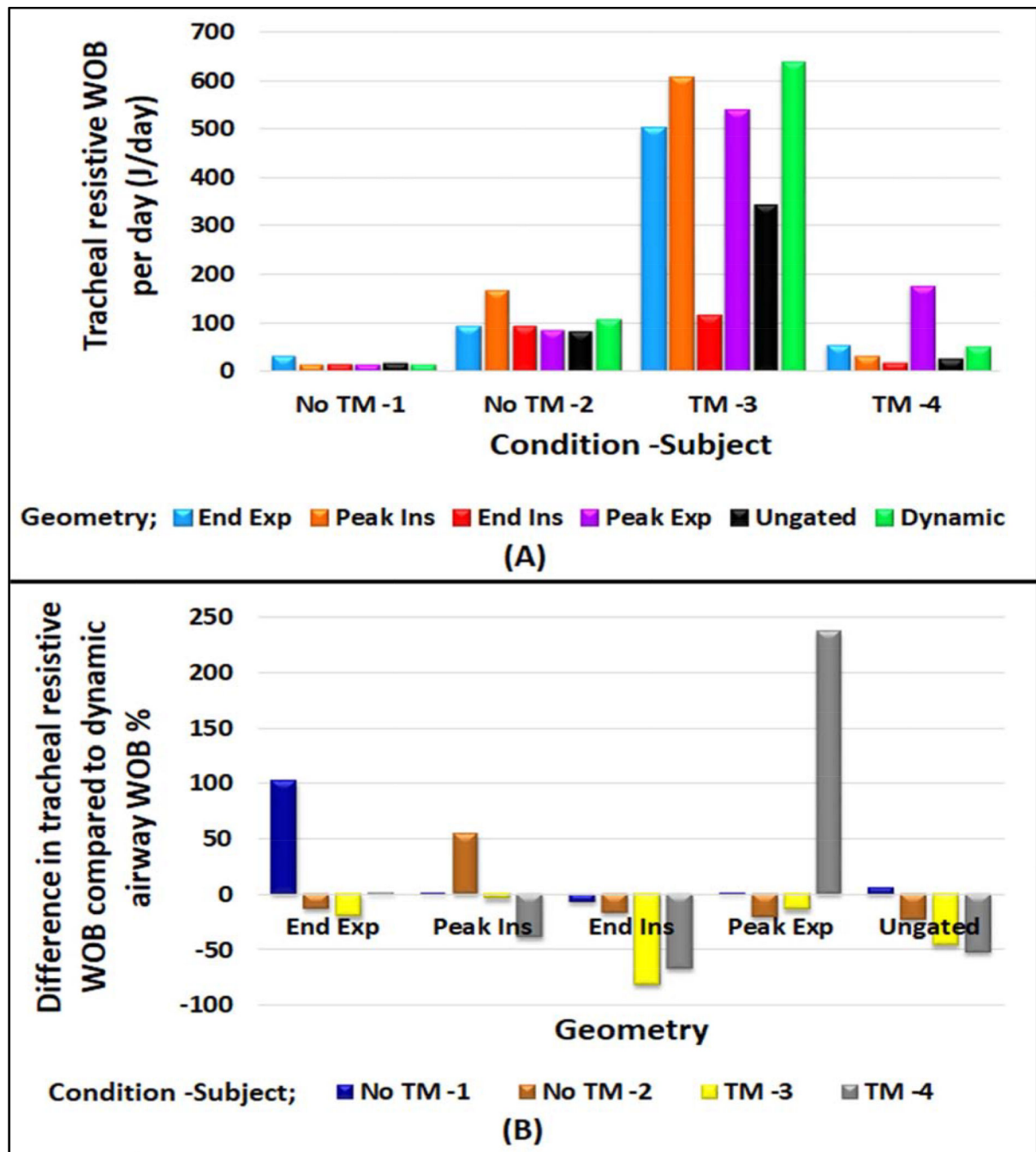


Figure 5.

Tracheal resistive WOB of each subject calculated per day using four static airway geometries obtained at different phases of breathing (blue – end expiration, orange - peak inspiration, red – end inspiration, and purple – peak expiration), ungated airway (black – ungated), and using the dynamic airway (green –dynamic) (A). The percentage differences of the tracheal resistive WOB compared to the dynamic airway at each phase of breathing and using the ungated airway in the four subjects (blue - Subject 1, brown – Subject 2, yellow – Subject 3, and gray – Subject 4) (B). As an example, in Subject 3 (with TM),

tracheal resistive WOB per day was 635.45 J using the dynamic airway and that value using end inspiration airway was 112.76 J (82% lower compared to the dynamic).

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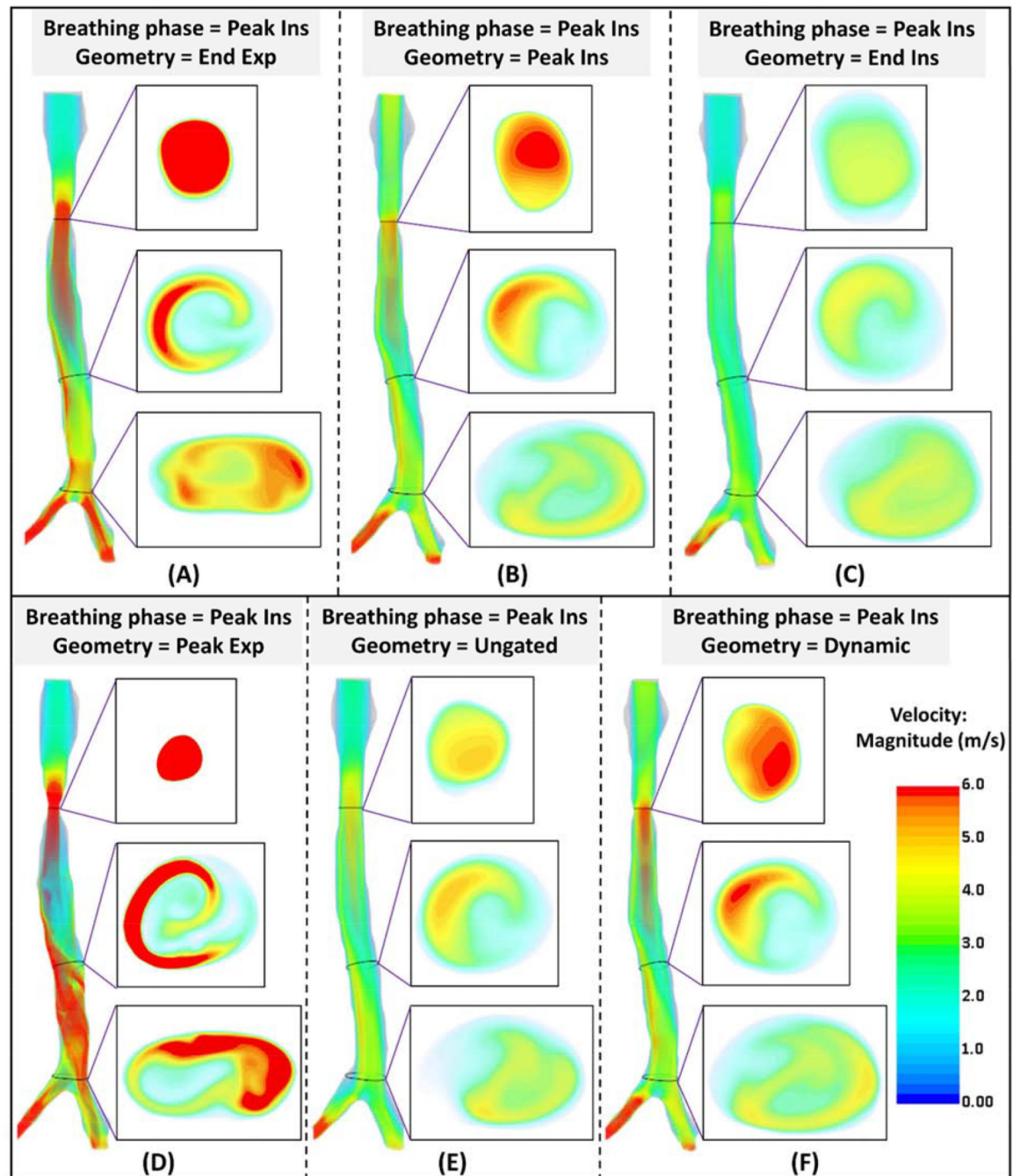


Figure 6.

Velocity distribution of the static airway geometries [end expiration (A), peak inspiration (B), end inspiration (C), peak expiration (D), and ungated (E)] and the dynamic airway (F) at peak inspiration airflow in subject 4. The velocity distribution of the glottis, middle of the trachea, and the carina cross-sections were shown for each simulation. Airflow velocity in the end expiration and peak expiration geometry was higher and lower in end inspiration and ungated geometry compared to the dynamic airway simulation.

Table 1.

UTE MR imaging parameters

Imaging parameter	Value
Image resolution	0.7 mm (3D isotropic)
Flip angle	5°
Field of view	18 cm
Echo time	200 μ s
Repetition time	5.2 ms
Number of projections	~200,000
Scan time	~16 minutes

Table 2.

Respiratory data for each subject

Subject (Condition)	Lung tidal volume (ml)	The ratio of right: left lung tidal volumes	Respiratory rate (breaths/min)
Subject 1 (without TM)	7.95	57 : 43	79
Subject 2 (without TM)	14.8	48 : 52	83
Subject 3 (with TM)	23.1	65 : 35	54
Subject 4 (with TM)	13.0	55 : 45	68